

Question	Answer	Marks
6(a)(i)	$I = 1.3 / 1.1$ $= 1.2 \text{ A}$	A1
6(a)(ii)	$v = I / nqA$ $= 1.2 / (8.5 \times 10^{28} \times 1.60 \times 10^{-19} \times 4.7 \times 10^{-7})$	C1
	$= 1.9 \times 10^{-4} \text{ m s}^{-1}$	A1
6(a)(iii)	$\rho = RA / L$	C1
	$= (1.1 \times 4.7 \times 10^{-7}) / 0.45$	C1
	$= 1.1 \times 10^{-6} \Omega \text{ m}$	A1
6(b)(i)	(Total) resistance decreases (and the potential difference stays the same)	M1
	(so the reading on the ammeter) increases	A1
6(b)(ii)	(The average drift speed will be) the same because the current is the same (in X).	B1

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